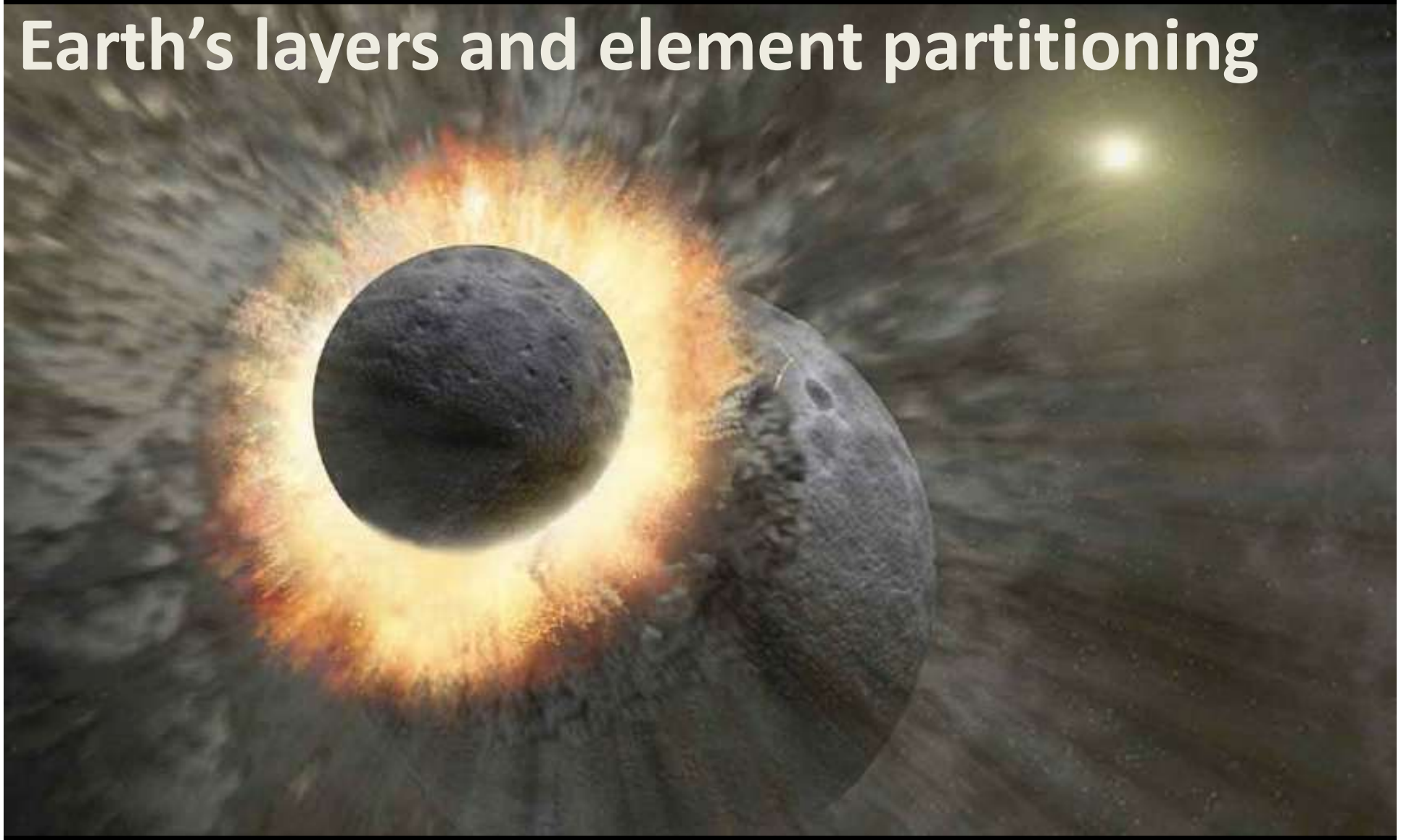
Chemistry of the Earth: Lecture 3

Earth's layers and element partitioning

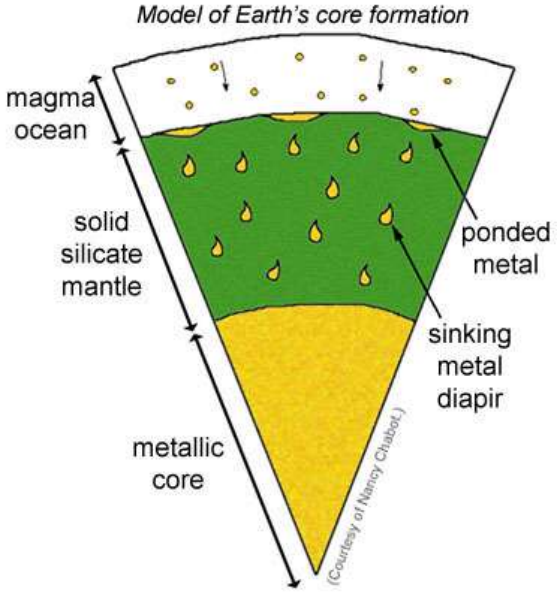


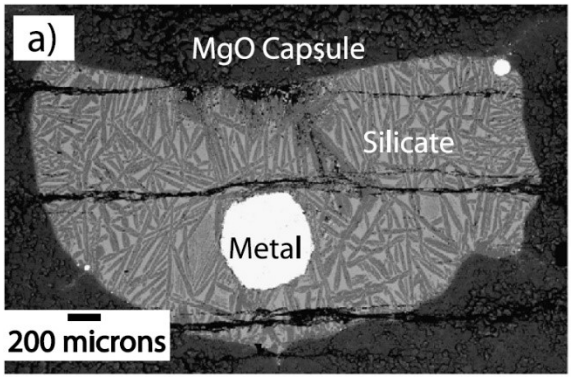
Part 1: Formation of the Earth's Core

IMAGE CREDIT: NASA/JPL-CALTECH/T. PYLE

OBJECTIVES OF THIS LECTURE

- Understand models for how the Earth's core formed
- Know why chemical elements are classified as siderophile, chalcophile, lithophile, atmophile (Goldschmidt's classification)
- Introduce the concept of element partitioning
- Investigate how the platinum group elements (PGEs) can be used to investigate geologic processes





	Fe 26	Co 27	Ni 28	Cu 29	Z 3
	Ru 44	Rh 45	Pd 46	Ag 47	C 4
	Os 76	Ir 77	Pt 78	Au 79	H 8

READING

Basic (minimum reading):

Grotzinger & Jordan	**Core formation: p. 603-604** **Mass extinctions: p. 247-249**

Recommended:

Misra (Recommended)	**Chapter 3.11 (Goldschmidt's classification)** **Chapter 12.2.3 (Core formation)**

Additional:

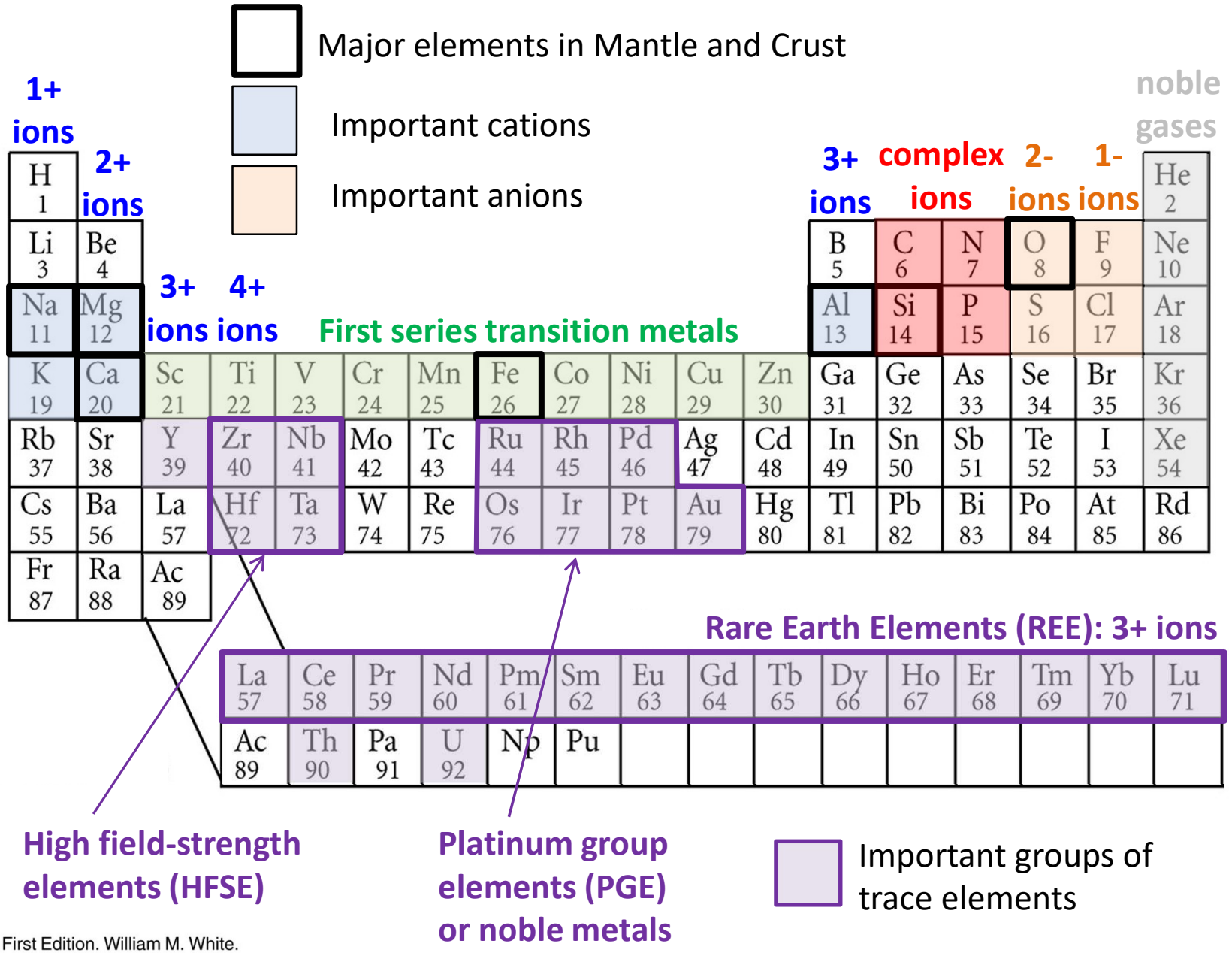
Marshak (Basic)	Chapter 1
Gill	p.215-220 (Goldschmidt, Ir, core formation)
McSween & Richardson	Chapter 2, p.23-24 (Goldschmidt)
White	Chapter 7.2.1 (Goldschmidt); Chapter 7.2.2.7 (PGEs)

Video

Collision that formed the Moon	https://www.youtube.com/watch?v=o2lRpiediP8

Geochemist's Periodic Table

See Lecture 2

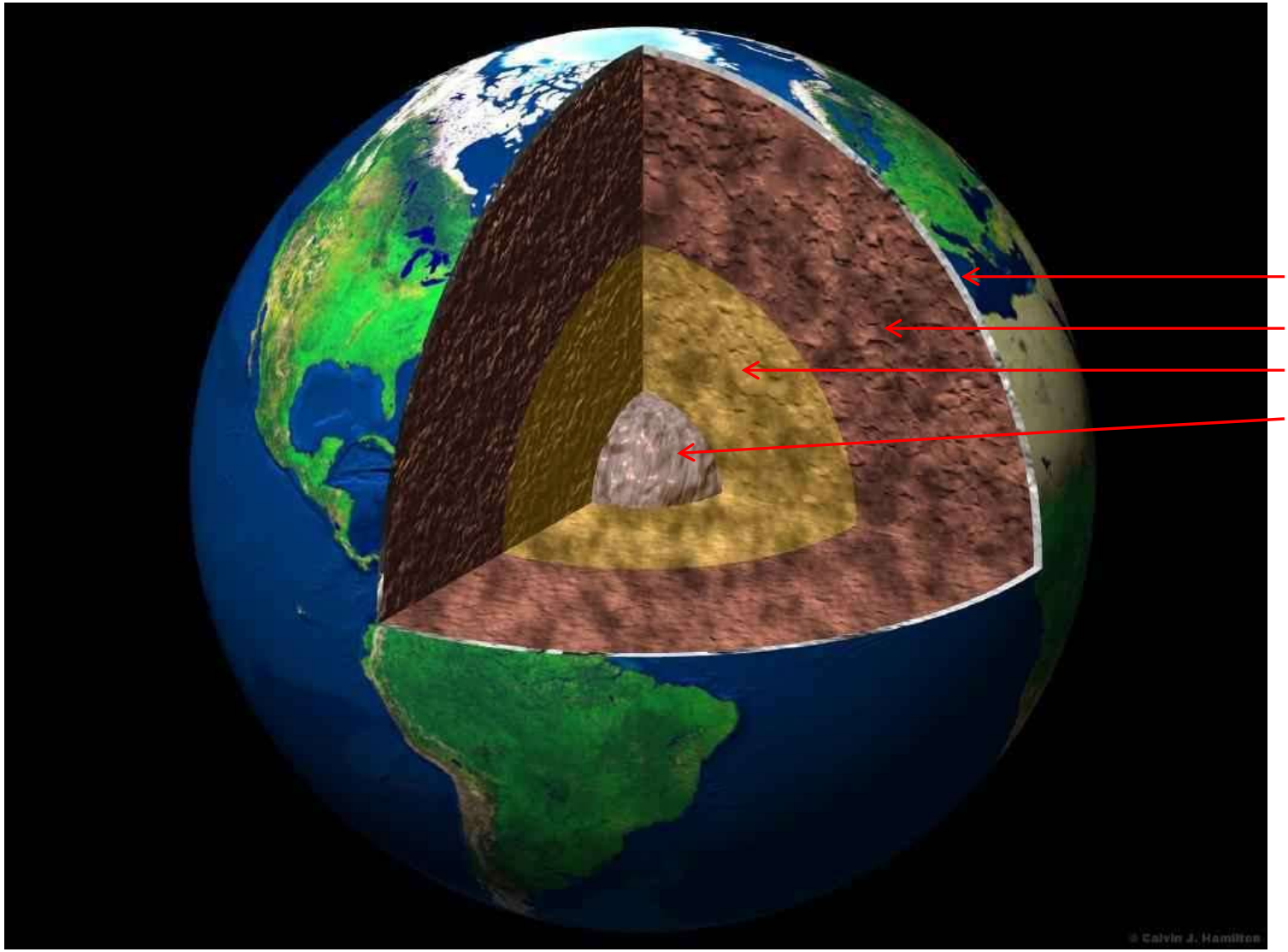


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EARTH'S LAYERS

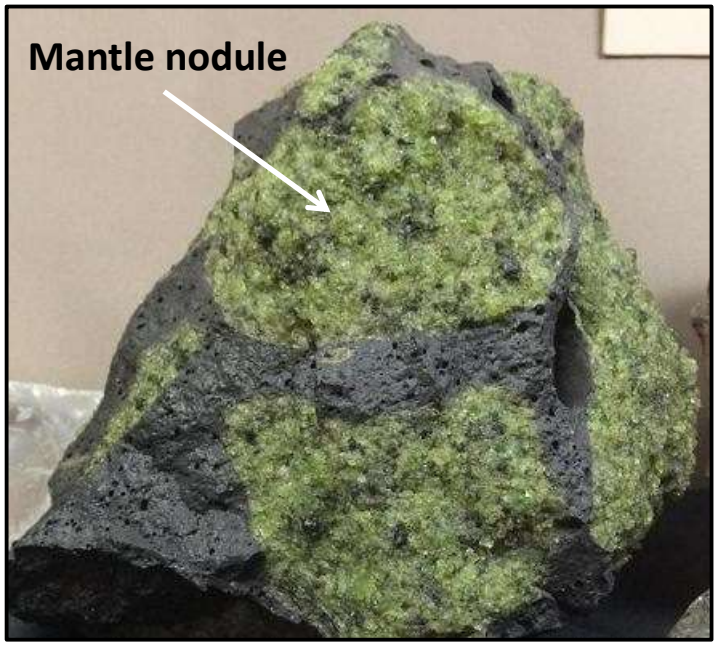
How and when did the core separate from the mantle?

What elements are concentrated in the core, and why?

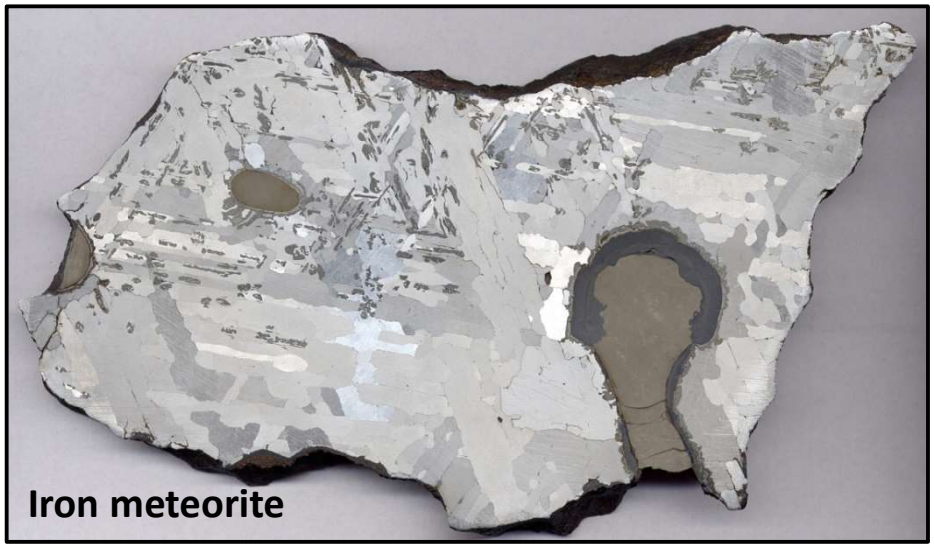


www.universetoday.com/wp-content/uploads/2008/10/earth_interior.jpg

MANTLE AND CORE: Rock and metal



http://peridot1.blogspot.co.uk/



Andrzej S. Pilski

- The Earth was hot when it formed.
- Heat sources include energy from the collisions of accretion, radioactive decay, gravitational energy released during core formation.
- Giant moon-forming impact was very energetic.
- Heating was sufficient to melt metallic iron and some (not all?) silicate.



https://en.wikipedia.org/wiki/Giant-impact_hypothesis

IMMISCIBLE LIQUIDS: Liquids that don't mix



http://chemistry.about.com/od/chemistryglossary/g/Immiscible-Definition.htm



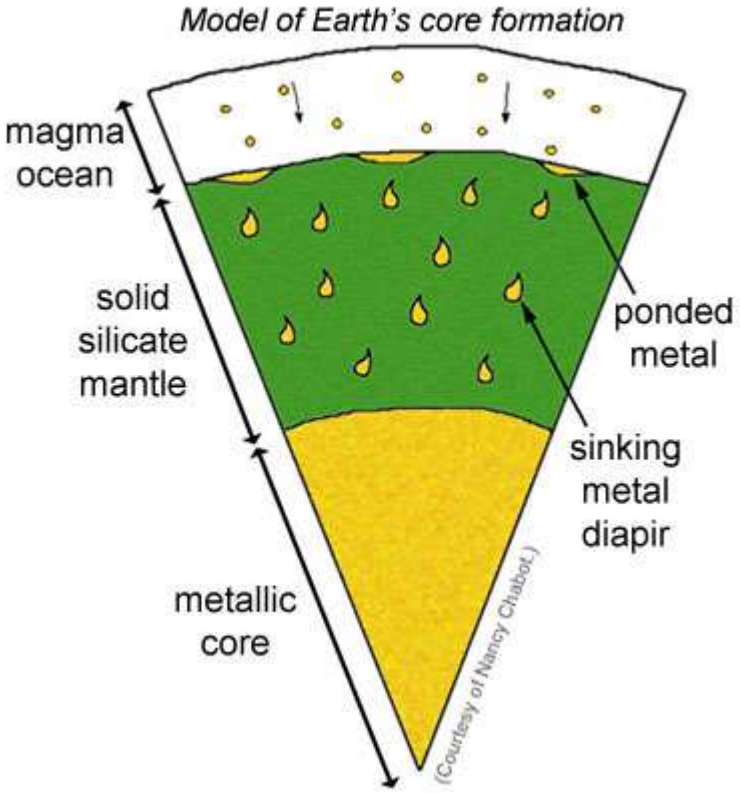
https://contentequalsmoney.com/wp-content/uploads/media/2013/09/lava-lamp.jpg

Two liquids with different bonding properties don't mix to form a single liquid. The denser of the two liquids sinks.

- Molten iron and molten silicate are immiscible liquids. They have different bonding – ionic bonding in silicate, metallic bonding in iron.
- Iron metal is denser and sinks.

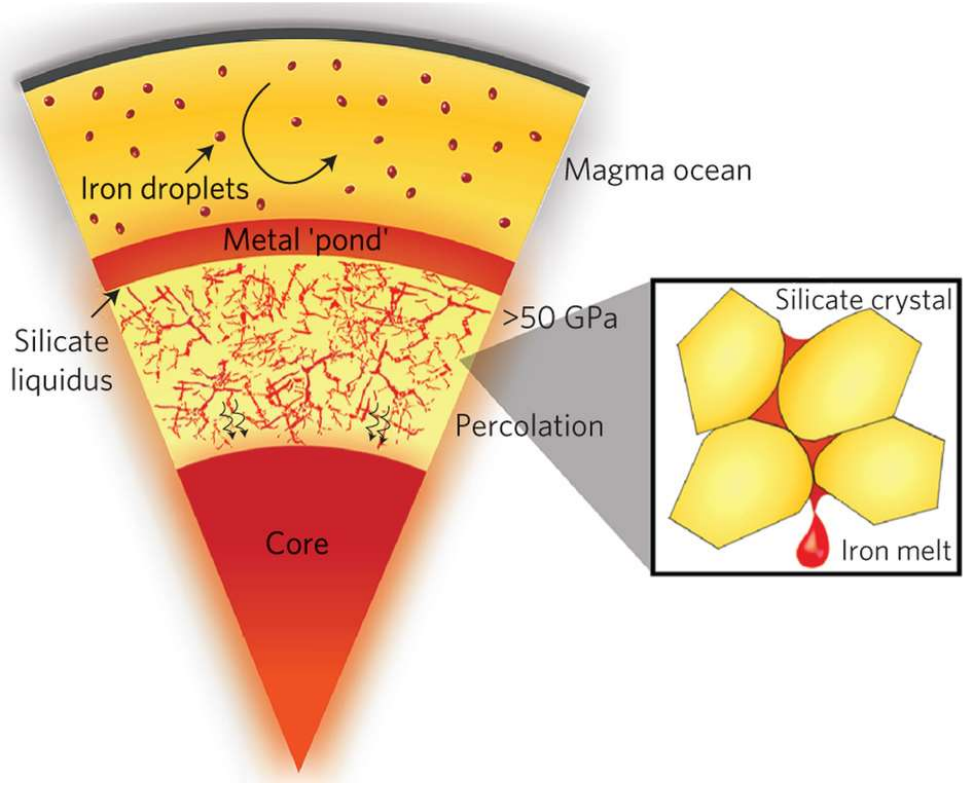
MAGMA OCEAN ON THE EARLY EARTH

Models for core formation include a “magma ocean” layer of molten silicate material.



http://www.psrd.hawaii.edu/July05/cobalt_and_nickel.html

This model has diapirs (pods) of metal sinking through the solid mantle

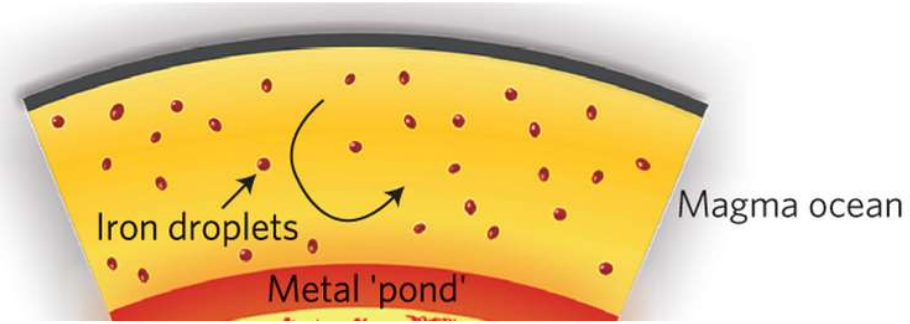


http://www.nature.com/ngeo/journal/v6/n11/pdf/ngeo1956.pdf

This model has liquid metal percolating down through spaces between silicate grains

- Measurements of isotopes such as ^{182}W (tungsten) show that the Earth’s core formed about 50 million years after formation of the Solar System

CORE FORMATION SEPARATED MANY ELEMENTS, NOT JUST IRON

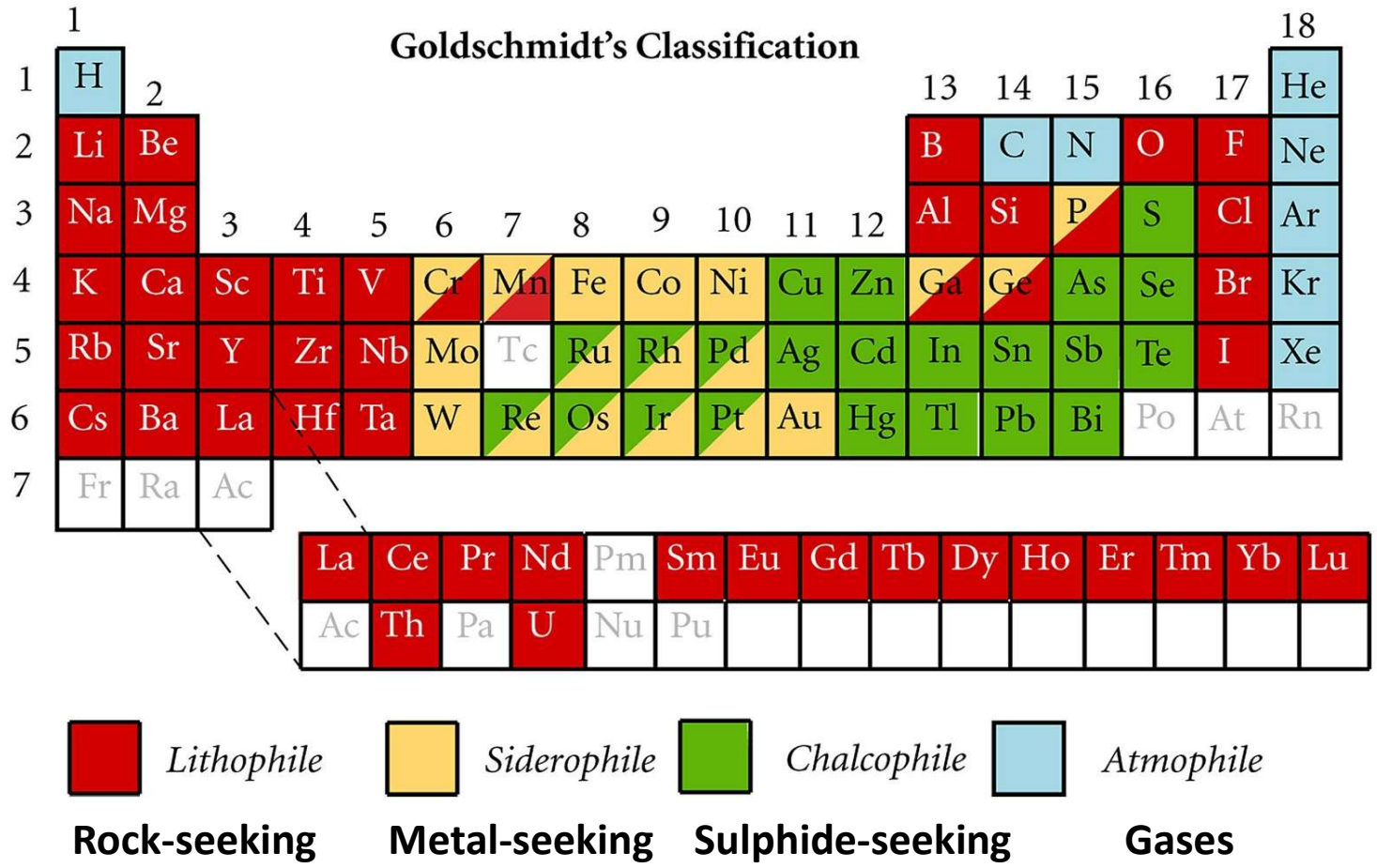


Core formation = Differentiation

Where iron metal and molten silicate are mixed, each element has a preference for which liquid it enters. Iron is split between the two liquids, as metal in the iron liquid and as ions (Fe^{2+}) in the silicate liquid .

SILICATE LIQUID	**IRON LIQUID**
Major elements Si, O, Mg, Fe, Ca, Al	Major elements Fe, Ni, S, Si, O
K, Rb, Cs, Ti, F, Cl, U, Th, REE etc.	Co, Ru, Pd, Ir, Pt, Au, W, etc.
LITHOPHILE: rock-seeking	**SIDEROPHILE**: iron-seeking

GOLDSCHMIDT'S CLASSIFICATION OF THE ELEMENTS

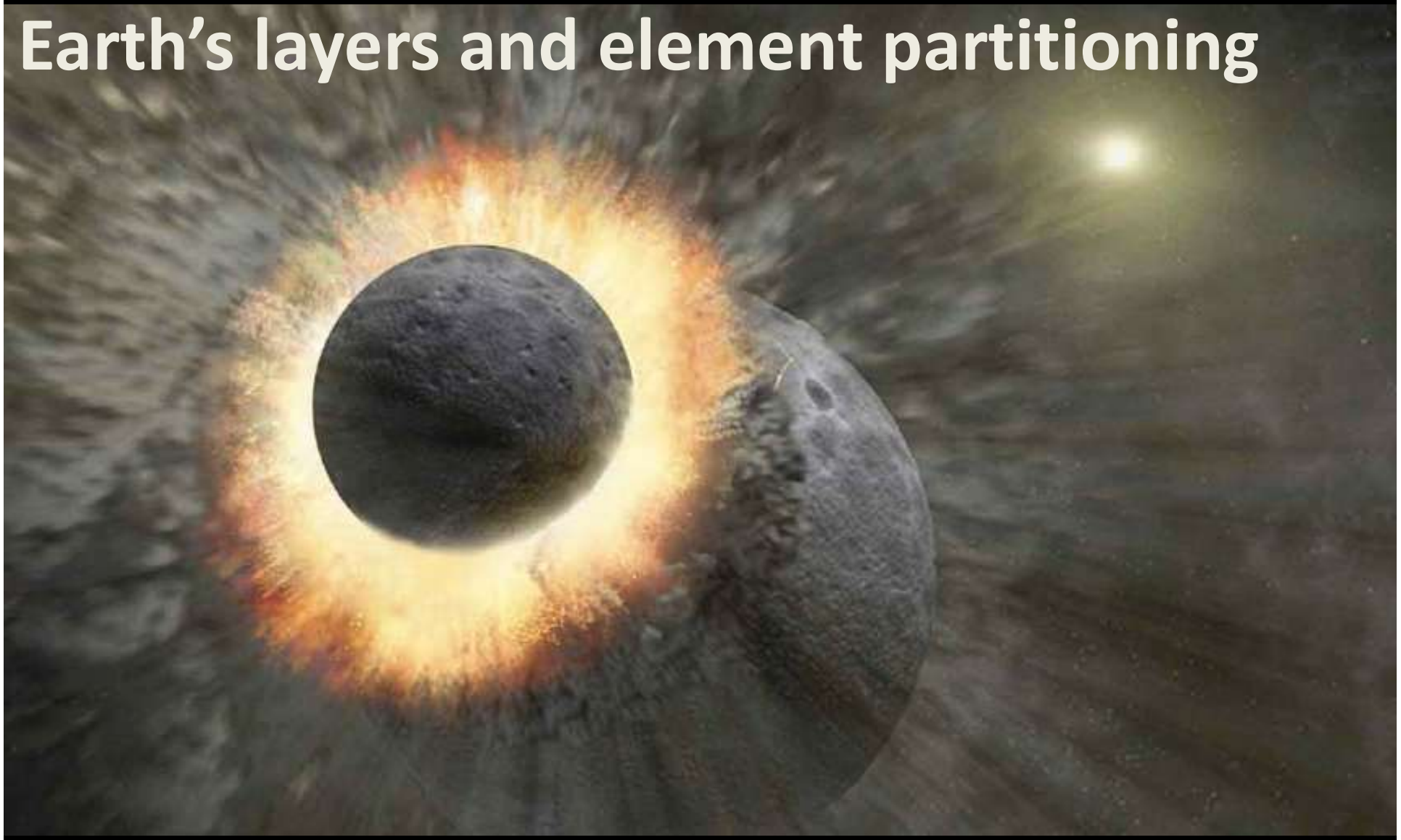


- Classification is based on the preference of the element for a silicate, metal or sulphide *liquid*. All three can coexist as immiscible liquids.
- Elements with two colours can occur in either of the two liquids.
- Chemical behaviour is related to the way an element bonds.

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Chemistry of the Earth: Lecture 3

Earth's layers and element partitioning



Part 2: Element Partitioning and the PGEs

IMAGE CREDIT: NASA/JPL-CALTECH/T. PYLE

PREFERENCE FOR ONE LIQUID – PARTITIONING

- Preference of an element for one of the liquids is called the *partitioning* behaviour of that element.
- For each element, the degree of preference between two liquids depends on the conditions - such as temperature, pressure, and amount of oxygen available.

We can measure the preference in terms of a ***partition coefficient***:

$$\text{Partition coefficient} = \frac{\text{Abundance in liquid 1}}{\text{Abundance in liquid 2}}$$

Abundances are measured as weight percent (wt%) values, or ppm, ppb etc.
Partition coefficients have no units.

Example:

$$\text{Metal / silicate partition coefficient for nickel} = \frac{\text{wt\% Ni in metal}}{\text{wt\% Ni in silicate}}$$

Partition coefficients (also known as distribution coefficients) are given the symbol D.

The example above is: $^{Ni}D_{met/sil}$

Partition coefficients are measured in experiments

Multi-anvil Press



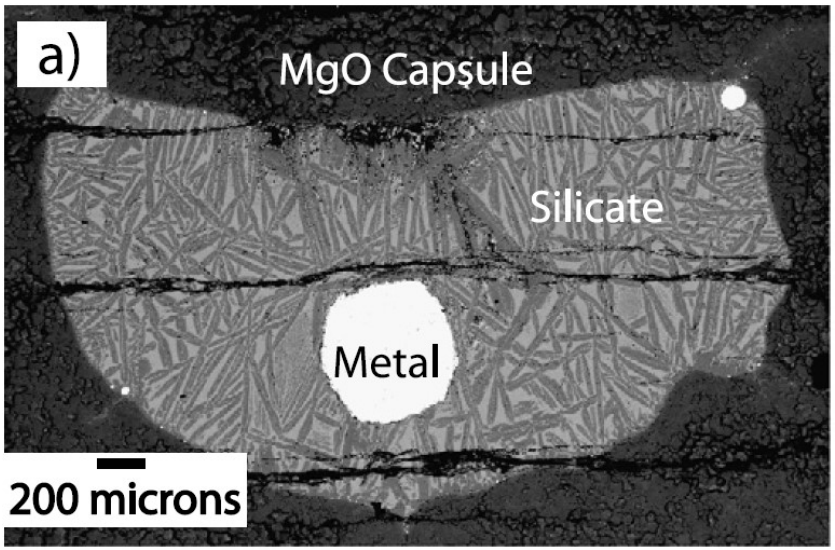
(Courtesy of Nancy Chabot.)

Right: High-pressure, high-temperature apparatus used by Chabot and her colleagues.

Below: Ceramic octahedral sample holder. Samples are small, which allows the equipment to attain high pressures.



http://www.psrd.hawaii.edu/July05/cobalt_and_nickel.html



Blanchard et al. (2015) Earth and Planetary Science Letters 427, 191–201

Experiment consisted of liquid silicate and liquid metal at high temperature and pressure.

Example: Metal contains 2.5 wt% Ni, Silicate contains 0.1 wt% Ni. $^{Ni}D_{met/sil} = 25$

- This means that the ratio of Ni in metal compared with Ni in silicate is 25:1.
- **Siderophile** elements have metal-silicate partition coefficients greater than 1.

Example: Ni content of the Earth's core

If we know the Ni content of the mantle, and the partition coefficient for Ni, we can estimate the Ni content of the core.

Table from Practical 1:

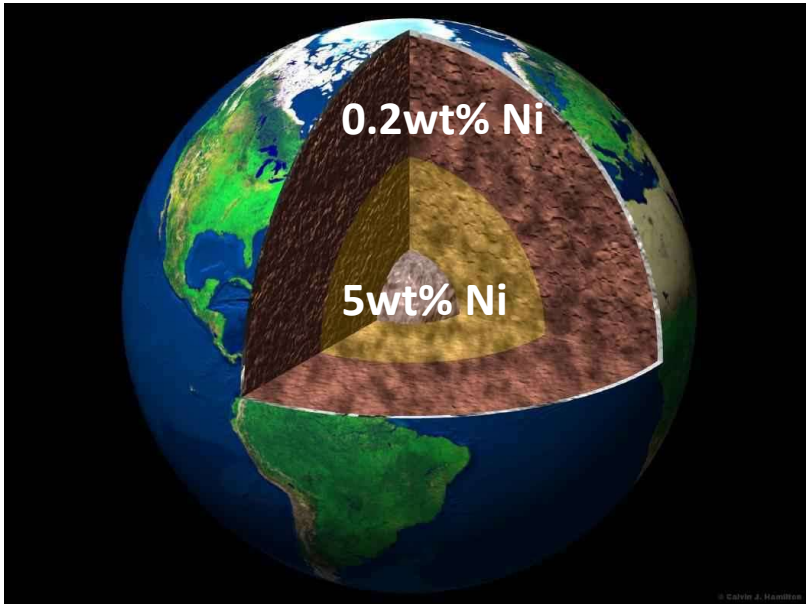
(abundances given in wt%)

Table 2: Major elements of the Mantle and Core		
Element	Mantle	Core
O	44.0	0
Na	0.27	0
Mg	22.8	0
Al	2.35	0
Si	21.0	6.0
S	0.03	1.9
Ca	2.53	0
Fe	6.26	85.5
Ni	0.20	5.2

If the Ni content of the mantle is 0.20 wt%

$^{Ni}D_{met/sil}$ = 25

Ni content of core is (0.2 * 25) = 5 wt%



Lithophile elements – preference for silicate

The metal / silicate partition coefficient for a lithophile element is less than 1.

Example: For manganese, Mn, $^{Mn}D_{met/sil} = 0.3$

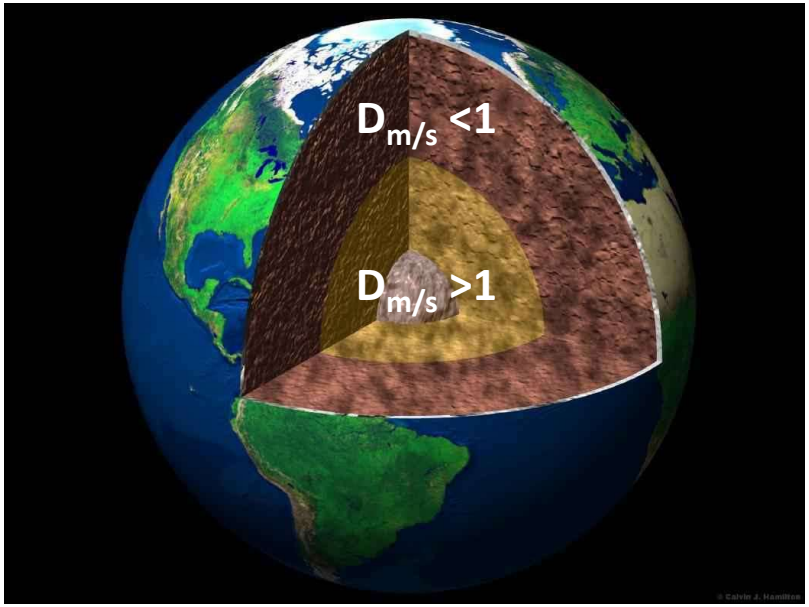
If there is 1045 ppm Mn in the mantle, how much Mn is in the core?

Amount of Mn in core / Amount of Mn in mantle = 0.3

Amount of Mn in core = 0.3 * Amount in mantle = 0.3 * 1045 = 313 ppm in the core

For a **siderophile** (metal-seeking) element, $D_{met/sil} > 1$

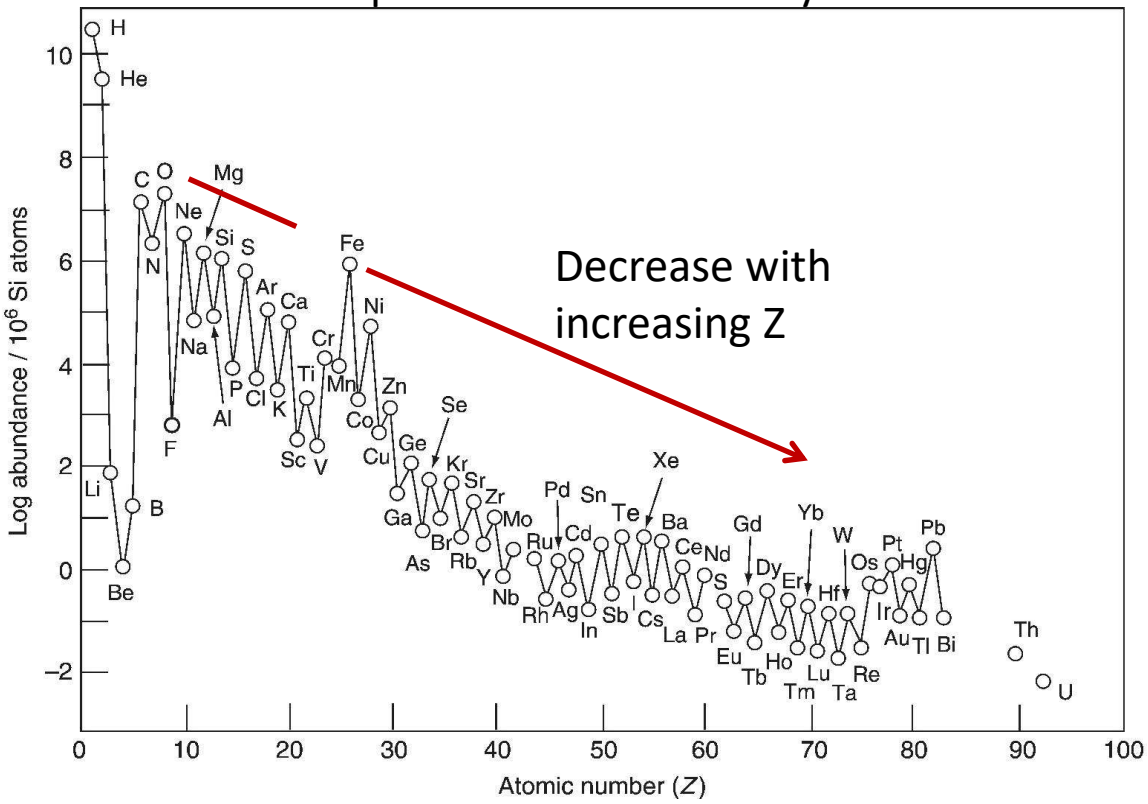
For a **lithophile** (silicate-seeking) element, $D_{met/sil} < 1$



THE POWER OF TRACE ELEMENTS

- Trace elements – elements present in low abundances.
- Depends on the material we are considering. An element may be a trace element in the core, but a major element in the mantle.
- In general, geochemically useful trace elements are heavier elements which are in low abundance throughout the Earth (remember nucleosynthesis).

Composition of the Solar System

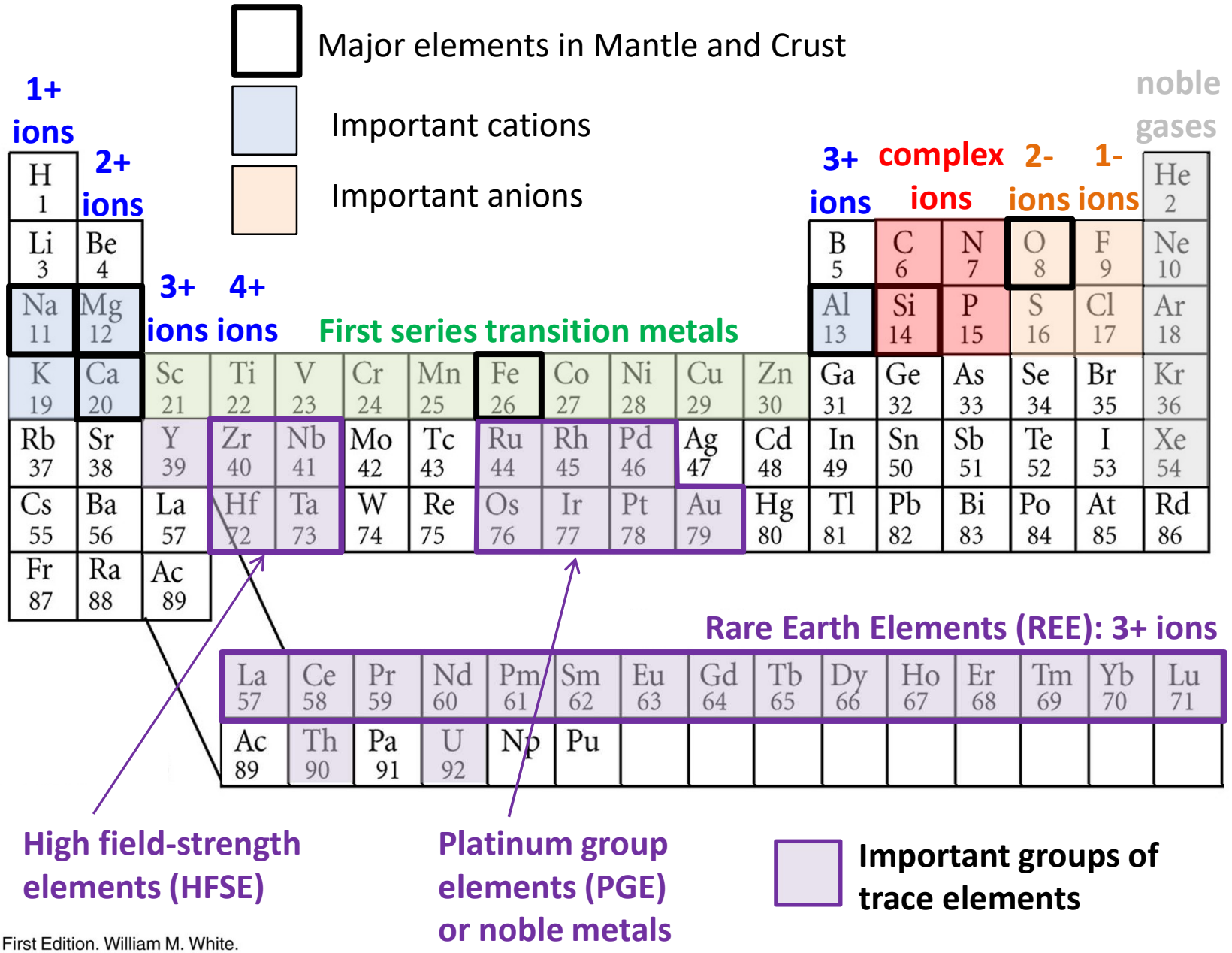


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See Lecture 1

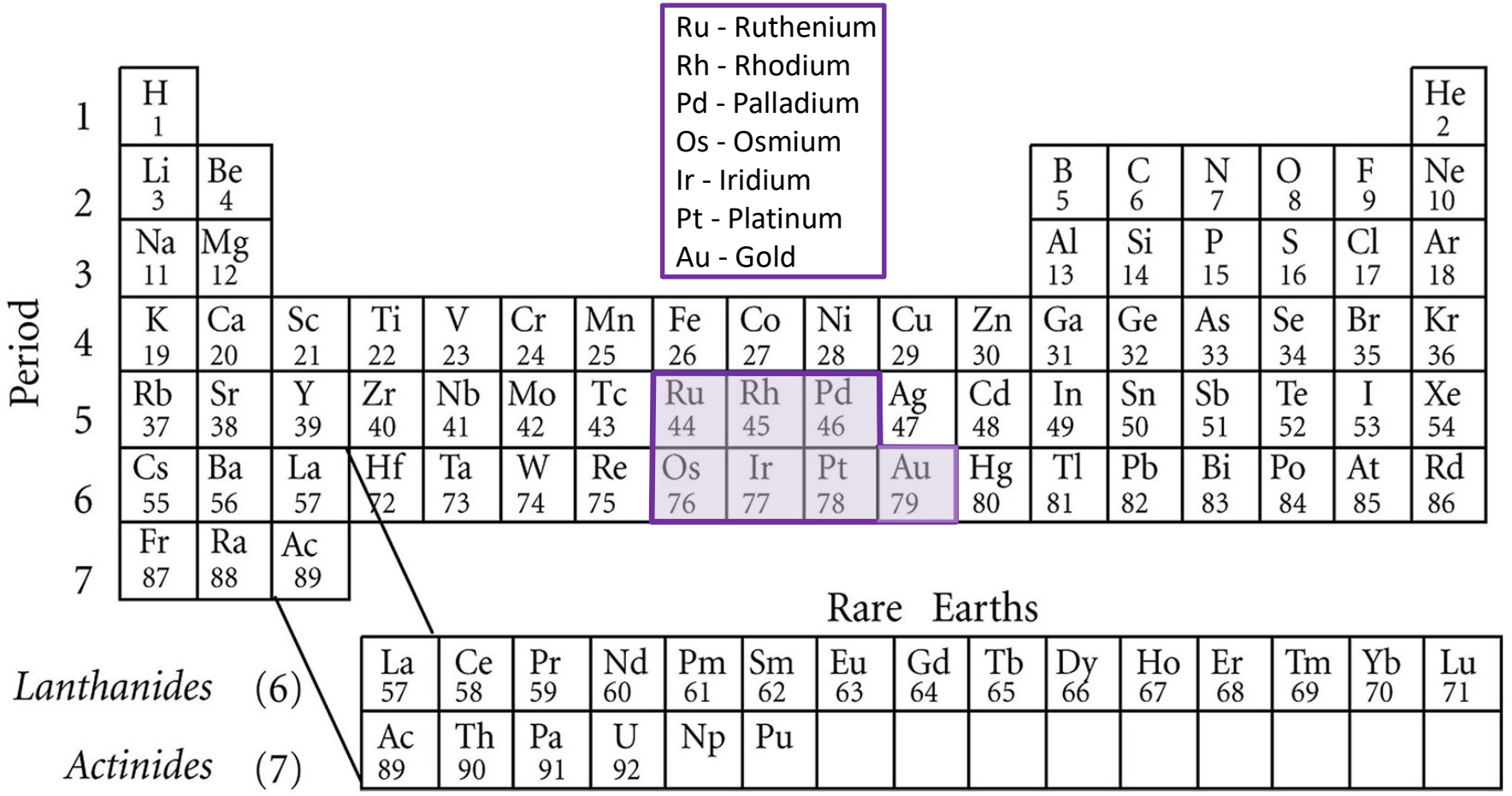
Geochemist's Periodic Table

See Lecture 2



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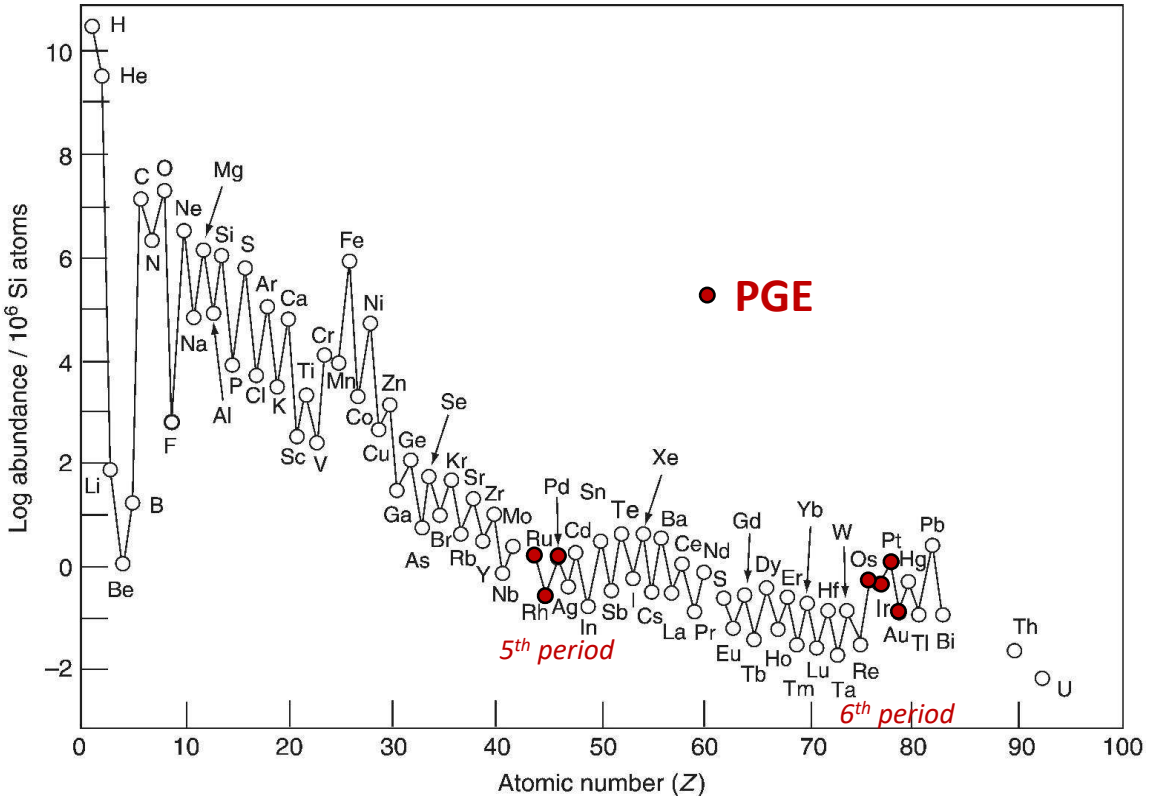
- We are going to focus on the **platinum-group elements** (PGEs)
- These elements are also known as **highly siderophile elements** (HSEs) – they have extremely strong affinities for metal compared with silicate.



Platinum group elements (PGE) or noble metals

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Abundances of PGEs in the Solar System and in the Earth



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Abundance in the Earth
(in parts per million – ppm)

Ru	1.3
Rh	0.24
Pd	1.0
Os	0.9
Ir	0.9
Pt	1.9
Au	0.16

HIGHLY SIDEROPHILE ELEMENTS

Metal / silicate partition coefficients for HSEs are very high.

Example: For gold (Au) $^{Au}D_{met/sil}$ is approximately 2,000.

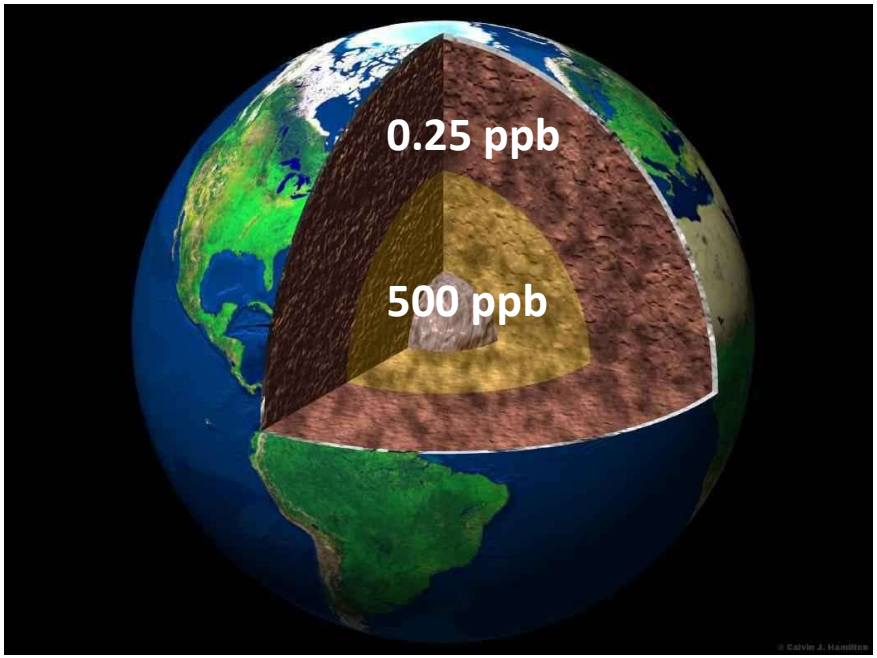
The bulk Earth contains approximately 160 ppb Au. (ppb = parts per billion)

~500 ppb Au in the core
~0.25 ppb Au in the mantle

Gold (Au)



Staffordcountymuseum.com



- **Most of the Earth's gold is in the core. The same is true for all the PGEs.**

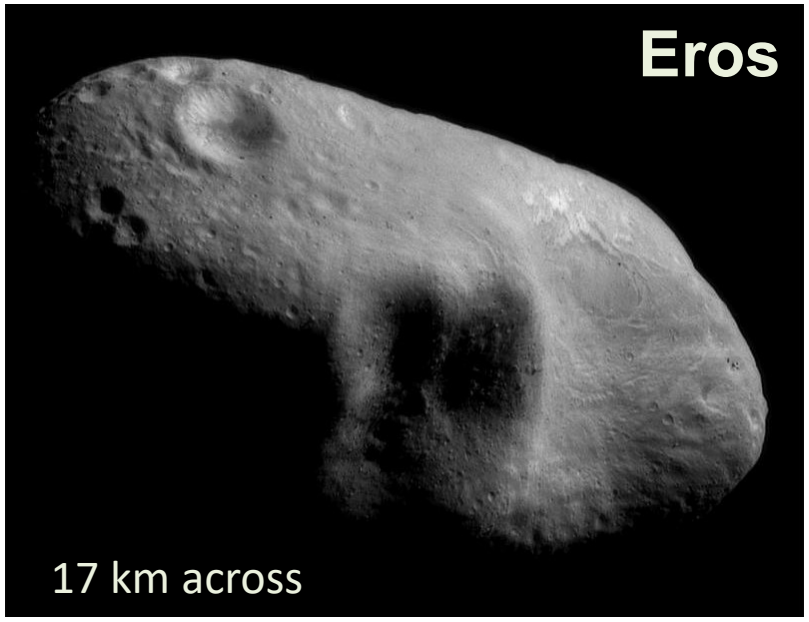
Is it worth mining asteroids for gold and other PGEs?



TABLE 15.4 Expected Lifetimes of Currently Known Ore Reserves (in Years)

Metal	World Resources	U.S. Resources
Iron	120	40
Aluminum	330	2
Copper	65	40
Lead	20	40
Zinc	30	25
Gold	30	20
Platinum	45	1
Nickel	75	<1
Cobalt	50	<1
Manganese	70	0
Chromium	75	0

Marshak p.584



- Mass of asteroid Eros is 7×10^{15} kg
- Eros is made of material like ordinary chondrite meteorites
- Abundance of gold in ordinary chondrites is about 200 parts per billion
- How much gold is there on Eros?

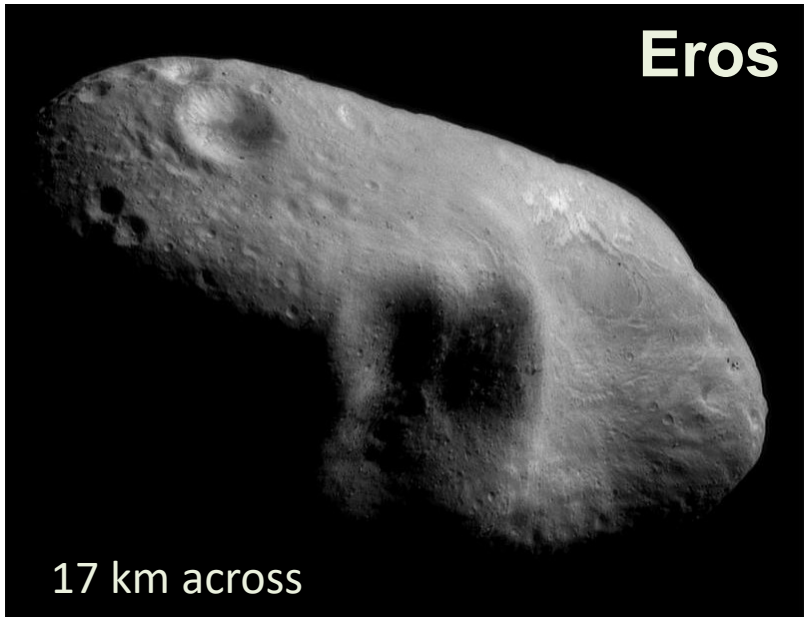
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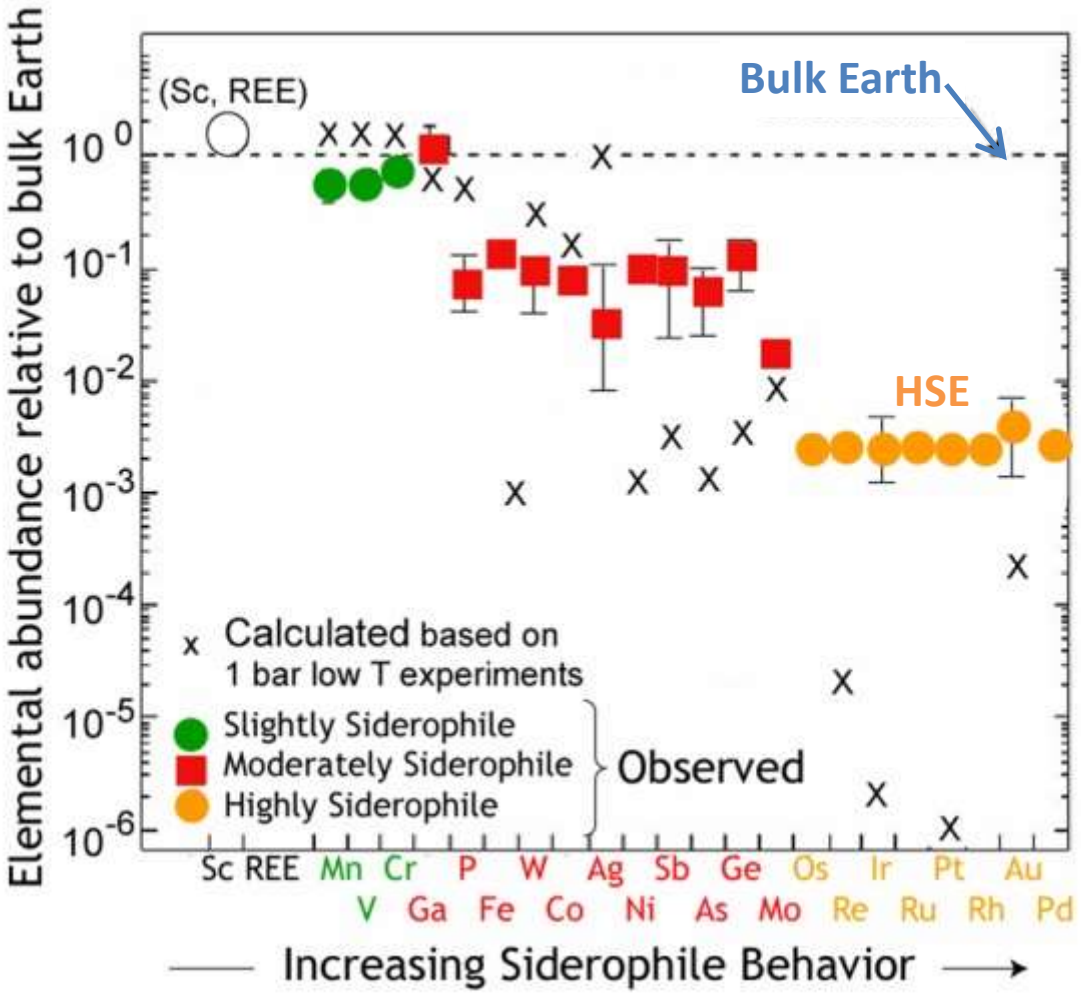
Marshak p.584



- Mass of asteroid Eros is 7×10^{15} kg
- Eros is made of material like ordinary chondrite meteorites
- Abundance of gold in ordinary chondrites is about 200 parts per billion
- Eros contains more than 1 billion kg of gold! (and all the other PGEs)

BUT: MANTLE ABUNDANCES OF HSE ARE ACTUALLY HIGHER THAN EXPECTED

Why? – a big question for geochemists.

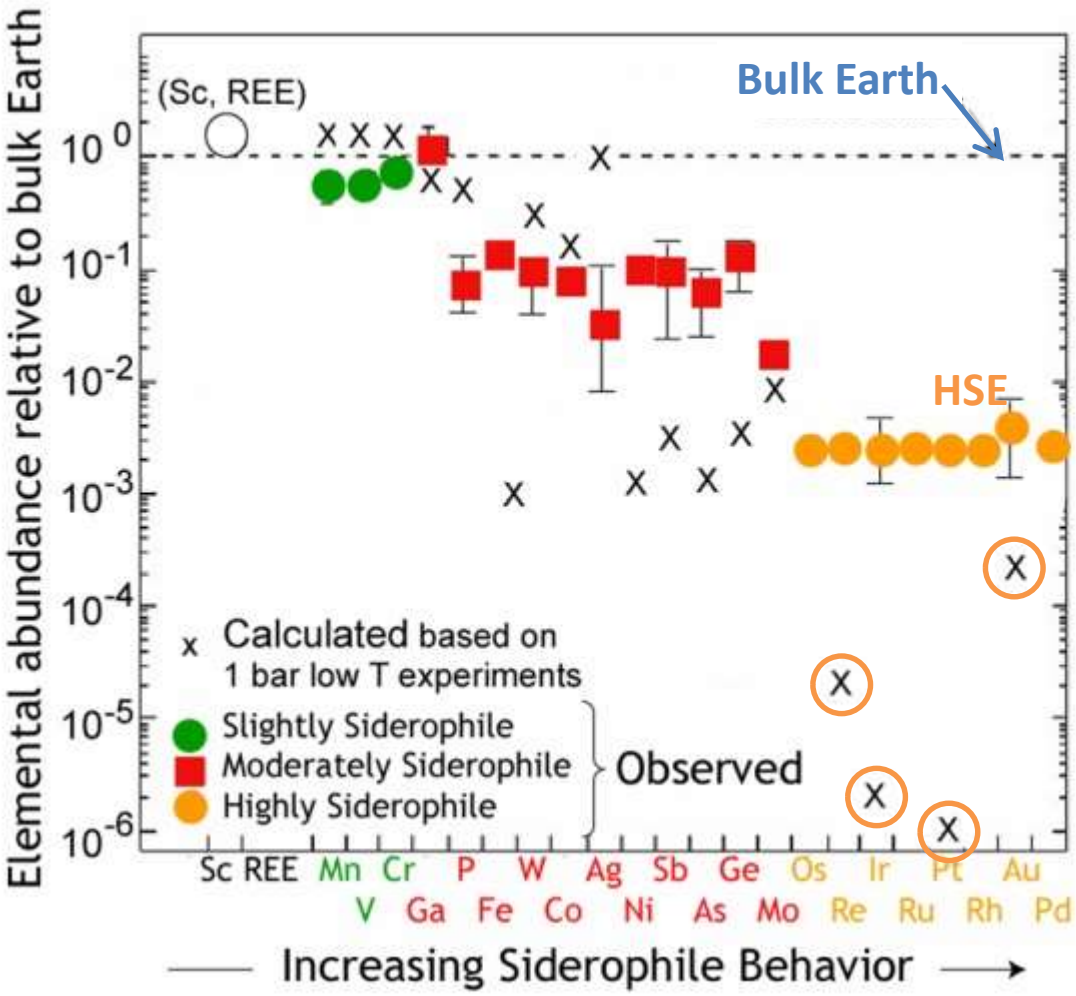


The observations –

The graph shows the abundance of siderophile elements in the mantle relative to the whole (bulk) Earth.

The x axis arranges the elements in order of increasing siderophile behaviour, i.e. their increasing affinity for metal.

Cottrell 2004: https://www.geol.umd.edu/home/richardwalker/research/siderophile.php



The graph also shows expected mantle abundances, calculated from experiments that have measured partition coefficients (crosses).

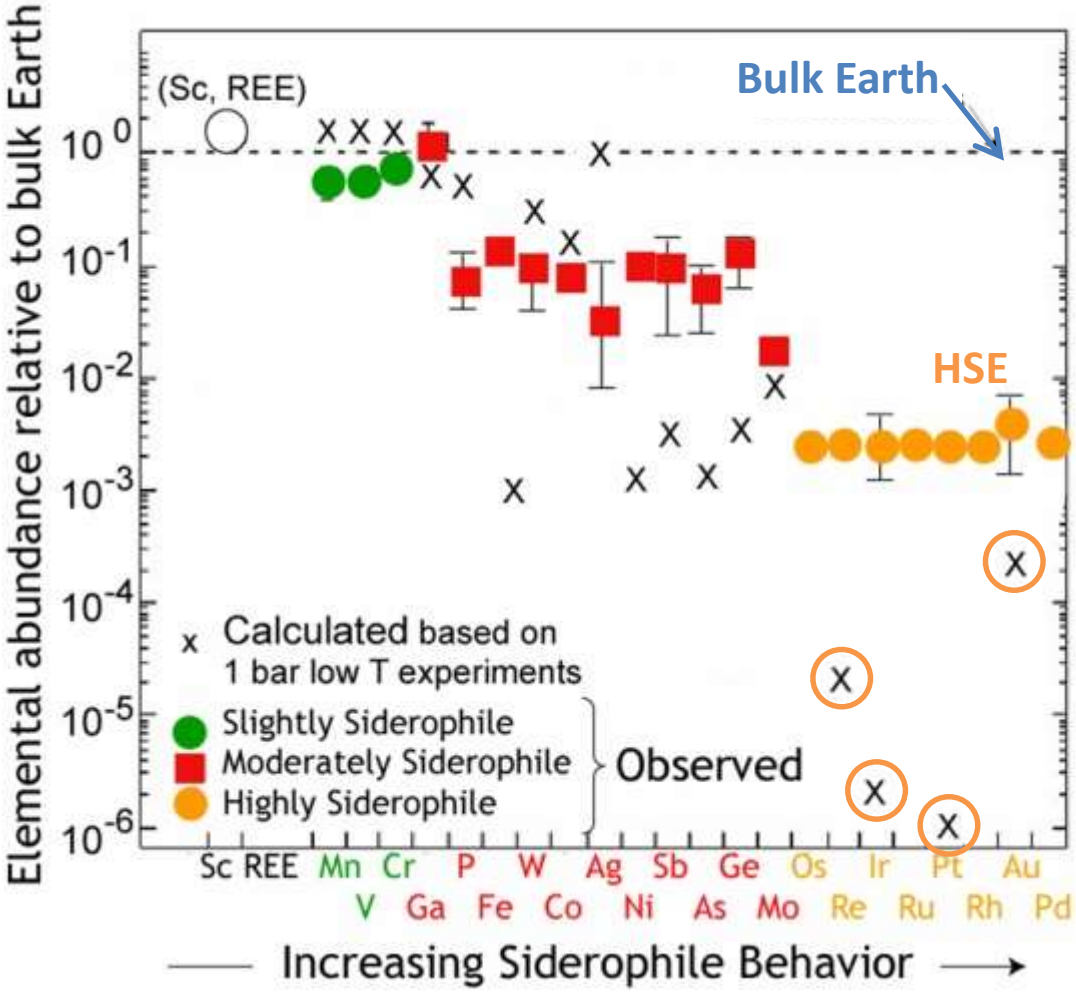
But there are two puzzling things about the graph:

- All the HSEs have similar abundances in the mantle (not expected).
- All the HSEs have higher abundances in the mantle than expected.

Cottrell 2004: https://www.geol.umd.edu/home/richardwalker/research/siderophile.php

Why are mantle abundances of HSEs higher than expected?

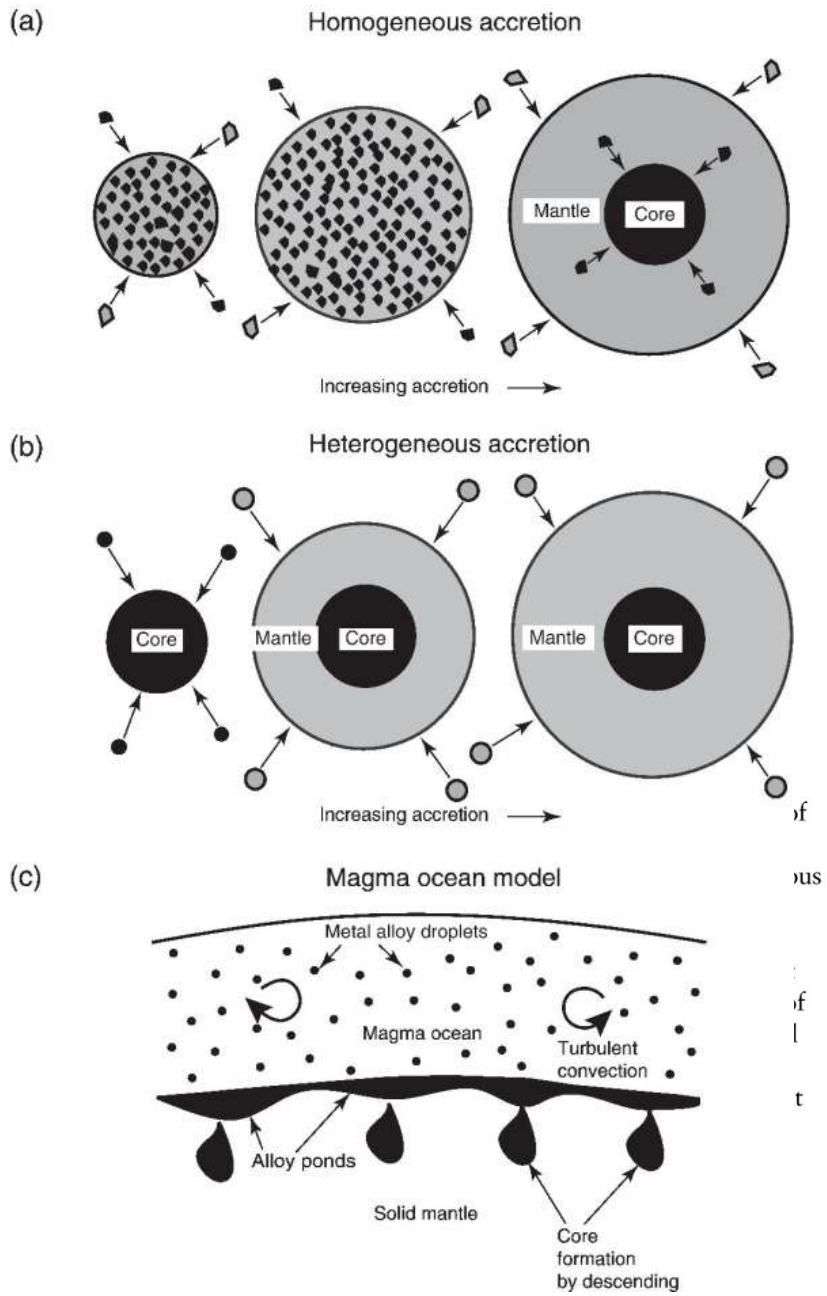
Various hypotheses have been proposed to explain the observations:



- Core separation was incomplete and some HSEs were trapped in the mantle.
- There was late accretion of chondritic material to the Earth, after core formation had taken place.
- Partition coefficients measured at one bar (one atmosphere) do not apply to the high pressures and temperatures of the deep mantle.
- More than one of the above processes.

(see Walker 2009, Chemie der Erde)

Cottrell 2004: https://www.geol.umd.edu/home/richardwalker/research/siderophile.php



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Models for the formation of Earth's core.
Misra, Fig. 12.11

a) Homogeneous accretion. Accretion of undifferentiated material followed by melting and core formation.

b) Heterogeneous accretion. The nature of what is accreting changes with time. Material that accretes late, after core formation, does not lose its siderophile elements to the core.

c) Magma ocean model. Metal separates from molten silicate in the magma ocean, sinks to the bottom of the magma ocean, then sinks as diapirs through the solid mantle into the growing core.

Chemistry of the Earth: Practical 3

Earth's layers and element partitioning

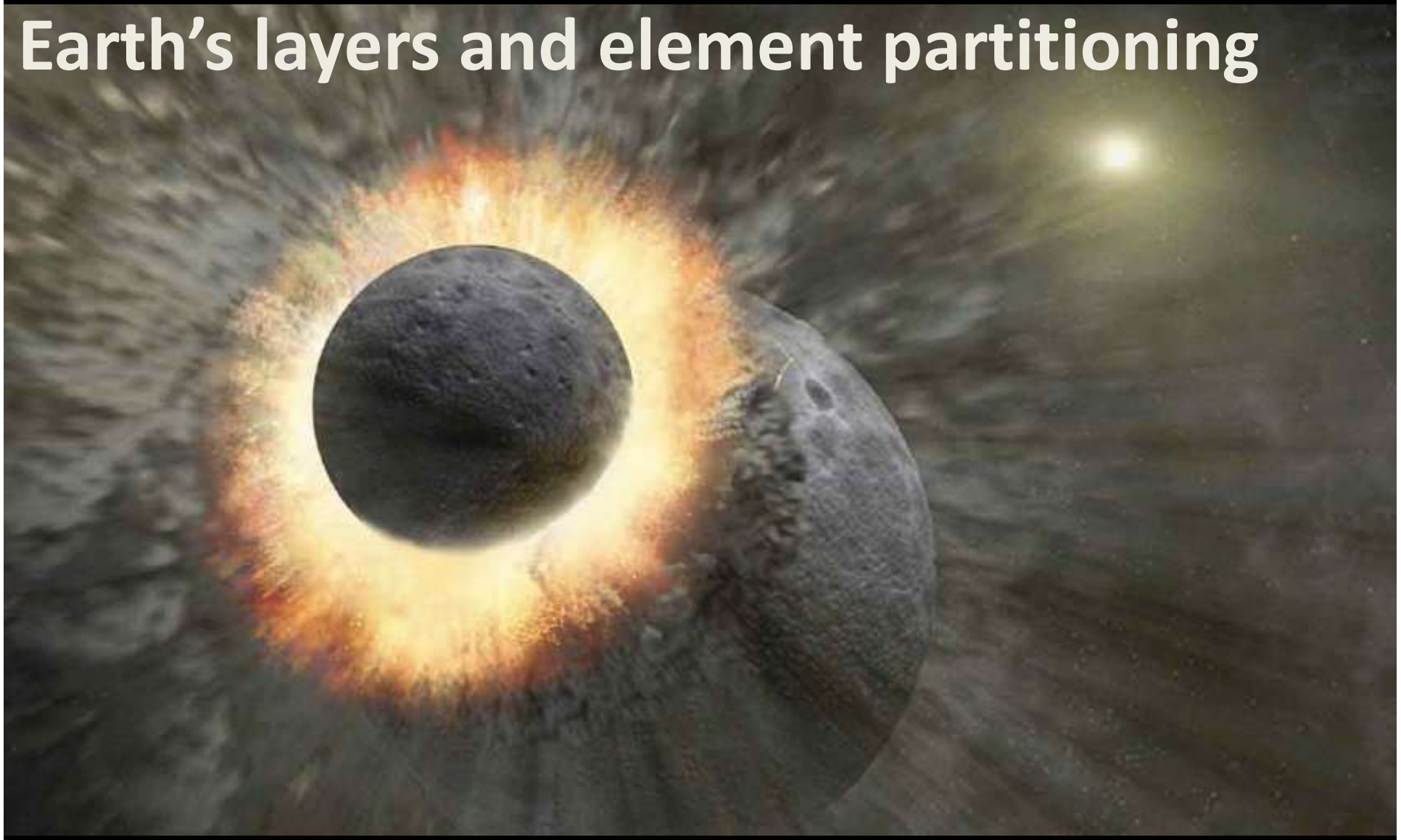


IMAGE CREDIT: NASA/JPL-CALTECH/T. PYLE

Practical 3:
Earth's layers and element partitioning

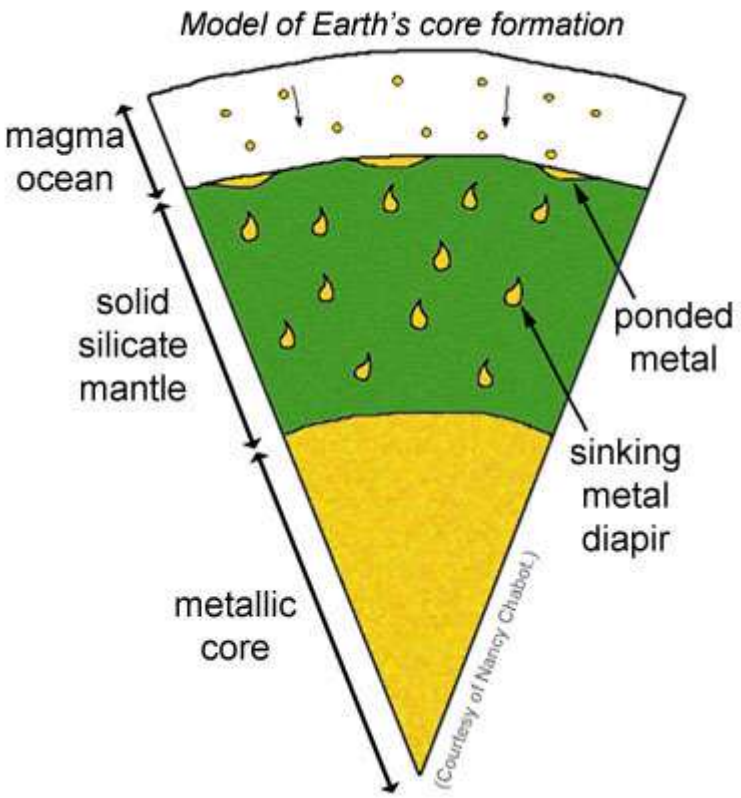
- Understand the concept of a partition coefficient and practice calculations.
- Investigate which elements are partitioned into the Earth's core, and why.
- Investigate the hypothesis that a large meteorite impact took place at the K/Pg boundary, using information from highly siderophile elements.

Core formation:

Magma Ocean:
Liquid silicate and liquid metal (iron)

Siderophile elements prefer the metal
Lithophile elements prefer the silicate

- The two liquids are immiscible
- Iron metal is denser and sinks to form the core
- Siderophile elements become concentrated in the core



PREFERENCE FOR ONE LIQUID – PARTITIONING

- Preference of an element for one of the liquids is called the *partitioning* behaviour of that element.
- For each element, the degree of preference between two liquids depends on the conditions - such as temperature, pressure, and amount of oxygen available.

We can measure the preference in terms of a ***partition coefficient***:

$$\text{Partition coefficient} = \frac{\text{Abundance in liquid 1}}{\text{Abundance in liquid 2}}$$

Abundances are measured as weight percent (wt%) values, or ppm, ppb etc.

Example:

$$\text{Metal / silicate partition coefficient for nickel} = \frac{\text{wt\% Ni in metal}}{\text{wt\% Ni in silicate}}$$

Partition coefficients (also known as distribution coefficients) are given the symbol D.

The example above is: $^{Ni}D_{met/sil}$

THE DEMISE OF THE DINOSAURS

Hypothesis:
The mass extinction of many species, including dinosaurs, at the end of the Cretaceous period (65 million years ago), can be attributed to the impact of a large meteor.

We will look at how highly siderophile elements (HSEs) play a role in supporting this hypothesis.



NASA



Getty images

K-Pg boundary at Stevns Klint, Denmark



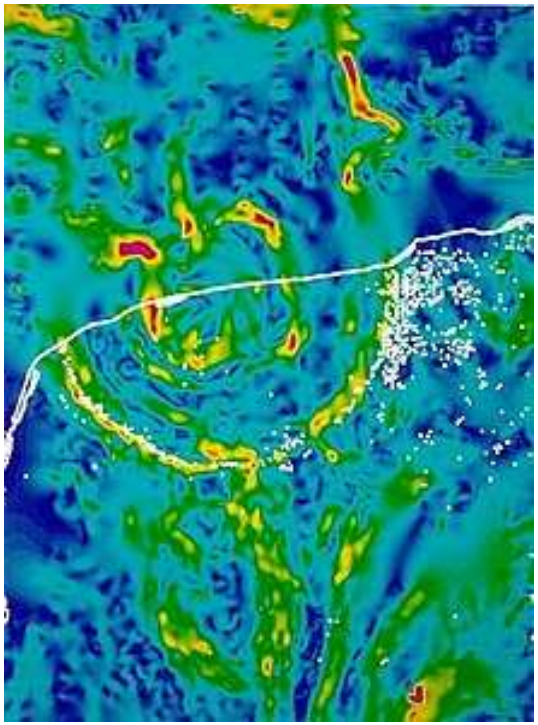
Stevns Klint in Denmark is a world-famous K-T boundary. It is composed of bryozoan limestone (upper part) and chalk (lower) which are both types of limestone. These layers are separated by a narrow layer of clayish material which formed 65 million years ago. The clay is very rich in iridium – a metal abundant in iron meteorites but rare on Earth. This layer formed as a result of meteorite explosion on the Yucatán Peninsula in Mexico. The event marks the end of the Mesozoic Era and is associated with the mass extinction that also ended the reign of dinosaurs. Dark gray spots are chert nodules and the boulders on the coast are also made of this rock type.

http://www.sandatlas.org/limestone/

Chicxulub impact structure, Mexico



Yucatán peninsula, Mexico



Gravity anomaly map of the Chicxulub impact structure. The coastline is shown as a white line. A striking series of concentric features reveals the location of the crater. White dots represent water-filled sinkholes (solution-collapse features common in the limestone rocks of the region) called *cenotes* after the Maya word *dzonot*. A dramatic ring of cenotes is associated with the largest peripheral gravity-gradient feature. The origin of the cenote ring remains uncertain, although the link to the underlying buried crater seems clear. *(Wikipedia)*

Impact crater is about 180 km in diameter.
Impactor was about 10-15 km across.